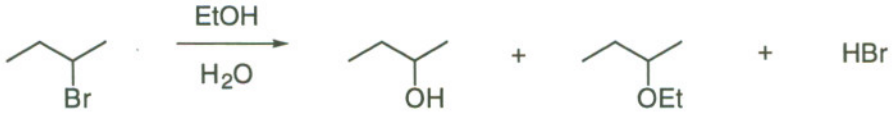
Given the following reaction:



2 **3**

A. (i) Write separate mechanisms for the formation of each solvolysis product **2** and **3** (showing all descriptive rate constants that could be experimentally determined) to express that the ratio of formation of the two products is under kinetic control.
(ii) Write separate mechanisms for the formation of each solvolysis product **2** and **3** (with rate constants that could be experimentally determined) that express the ratio of the two products is under thermodynamic control.

B. Give the rate law for each reaction from part A.

C. What determines which product would be favored under kinetic control? Express this in terms of relative rate constants from your mechanism.

D. What determines which product would be favored under thermodynamic control?

E. What experimental parameter would be altered, and how, in order to favor kinetic control in the reaction?

F. How would you favor the thermodynamic product?

G. What would be the difference between the reaction if run under neutral conditions versus acidic conditions?